Generative Modelling Using Generative Adversarial Networks Across Multiple Domains

1. Introduction

Generative modelling has become a central area in modern data science, enabling the synthesis of realistic data for applications such as simulation, data augmentation, and anomaly detection. Among the various generative approaches, **Generative Adversarial Networks (GANs)** have emerged as one of the most influential frameworks due to their ability to learn complex, high-dimensional data distributions without explicitly defining likelihood functions.

GANs are particularly effective in domains where data exhibits structured patterns, such as images. However, their performance is highly dependent on the nature of the data, the chosen architecture, and the evaluation strategy. This sensitivity makes GANs both powerful and challenging to apply across different domains.

This study investigates GAN performance across four distinct settings:

- controlled synthetic distributions
- medical image generation (BloodMNIST)
- tabular network traffic generation (CICIDS 2017)
- creative sketch generation (QuickDraw)

Rather than focusing solely on implementation, the primary objective is to **critically evaluate how GAN behaviour varies across these domains**, thereby demonstrating a deeper understanding of generative modelling principles. In particular, this study examines how differences in data structure influence the ability of GANs to learn meaningful representations.

While GANs are well established in image-based applications, their performance in non-visual domains such as tabular data and abstract sketches remains less consistent. This study therefore adopts a comparative, cross-domain approach, enabling a systematic analysis of how **data representation, model architecture, and evaluation methodology jointly impact generative performance**.

2. Methodology

Generative Adversarial Networks (GANs), first introduced by Ian Goodfellow et al. (2014), consist of two neural networks trained in opposition: a generator and a discriminator. The generator learns a mapping from a latent noise distribution to the data space, while the discriminator attempts to distinguish between real samples and those produced by the generator.

The training process is formulated as a minimax optimisation problem, in which the generator aims to maximise the probability of the discriminator misclassifying generated samples, while the discriminator seeks to correctly identify real and synthetic inputs. Through this adversarial interaction, the model progressively improves, allowing the generator to approximate the underlying data distribution. Unlike traditional generative models, GANs do not explicitly estimate likelihood functions; instead, they learn implicitly through this competitive process.

2.1 Architecture Selection and Justification

The choice of architecture in this study is guided by the structural characteristics of the data. For low-dimensional synthetic data and tabular datasets, **fully connected neural networks** are employed, as these data types do not exhibit spatial structure and rely primarily on feature-level relationships.

In contrast, for image-based tasks such as BloodMNIST and QuickDraw, a **Deep Convolutional GAN (DCGAN)** architecture is used, following the framework proposed by Alec Radford et al. (2015). Convolutional layers are particularly effective in these contexts because they exploit spatial locality, enabling the model to capture patterns such as edges, textures, and shapes.

This architectural distinction reflects an important principle in deep learning:
'the inductive bias of the model must align with the underlying data representation'.
By tailoring the architecture to the data type, the model is better equipped to learn meaningful features, thereby improving generative performance and training stability.

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3. Part 1: Building and Understanding GANs from Scratch

3.1 Objective and Experimental Design

The objective of this section is to develop a practical understanding of Generative Adversarial Networks (GANs) by implementing them from scratch using PyTorch on synthetic two-dimensional data. Synthetic datasets provide a controlled setting for analysing model behaviour without the complexity of real-world data.

A baseline GAN is first trained on a sine-wave distribution to evaluate its ability to learn smooth, continuous patterns. The model is then applied to a more complex 2D spiral distribution, introducing nonlinear structure and increased geometric complexity. Architectural modifications, including changes to network depth and activation functions, are also explored to assess their impact on generative performance and training stability.

3.2 Results and Observations

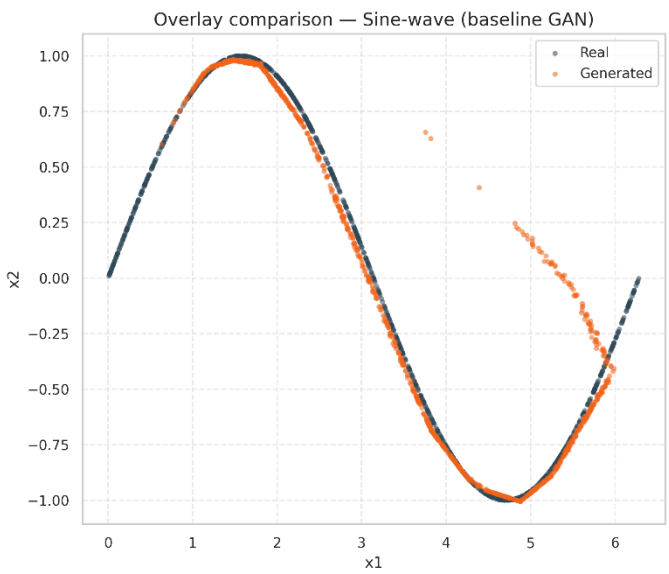

Figure 1: *Comparison of real and generated samples for the sine-wave distribution.*

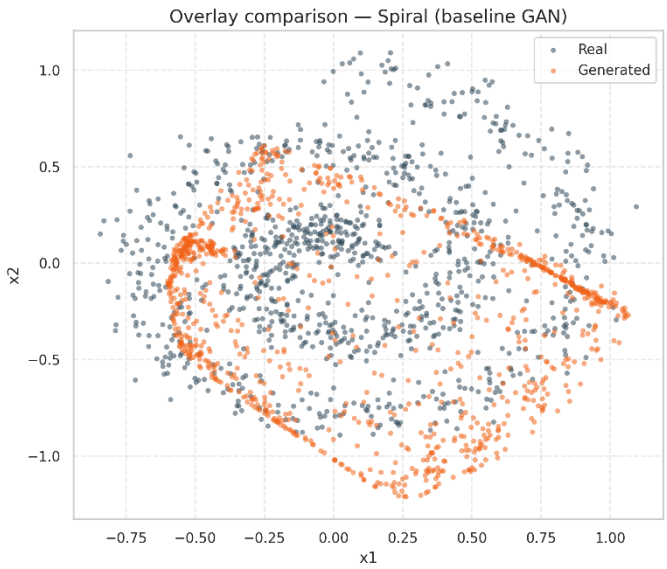

Figure 2: *Real and generated samples for the spiral distribution.*

The baseline GAN successfully learns the sine-wave distribution, with generated samples closely aligning with the true curve. This indicates that the model can capture smooth, continuous patterns when the underlying data manifold is relatively simple.

However, when applied to the spiral dataset, performance degrades noticeably as shown in Figure 2. While the generator is able to approximate the central region of the spiral, it fails to accurately reproduce the outer structure, resulting in incomplete and distorted representations. This suggests that the model struggles to capture global geometric properties in more complex distributions.

Architectural modifications produce mixed results. In some cases, deeper networks improve visual alignment with the target distribution, while in others they introduce instability, reflecting the sensitivity of GAN training to model configuration.

3.3 Critical Analysis

dataset_name	model_name	mean_l2_distance	std_l2_distance	training_time_seconds		
sine_wave	baseline_gan	0.3886	0.2284	102.74		
sine_wave	improved_gan	0.3735	0.5706	109.93		
spiral	baseline_gan	0.2738	0.0951	105.34		
spiral	improved_gan	0.2844	0.2328	108.07		

Table 1: *Quantitative comparison of baseline and modified GAN architectures using L2 distance and training time metrics.*

A key finding from this experiment is that quantitative metrics such as L2 distance do not consistently reflect perceptual quality. In several instances, models with higher L2 distance produced visually more accurate samples, highlighting a fundamental limitation of distance-based evaluation: such metrics fail to capture the geometric and structural properties of the data distribution.

As a result, evaluation must extend beyond numerical measures to include visual inspection and qualitative assessment. This is particularly important in generative modelling, where the objective is not merely to minimise error, but to reproduce the underlying structure of the data.

The difficulty in modelling the spiral distribution further illustrates a limitation of standard GANs: their reliance on implicit learning without explicit constraints on global structure. Consequently, complex nonlinear manifolds that require consistent representation across the data space are not fully captured. This reinforces the need for multi-faceted evaluation strategies when assessing generative models.

3.4 Insight

The results demonstrate that GANs perform effectively when the data distribution is smooth and continuous, as observed in the sine-wave experiment. However, performance deteriorates as structural complexity increases, particularly for distributions that require maintaining global geometric consistency.

This limitation arises because standard GANs learn distributions implicitly without enforcing constraints on the overall structure of the data manifold. As a result, the model can capture local patterns but struggles to maintain coherence across the entire distribution, especially in regions with sparse training samples.

Furthermore, the training dynamics reflect the inherent instability of GAN optimisation. The oscillatory behaviour between the generator and discriminator indicates an unstable equilibrium, where improvements in one network can temporarily degrade the performance of the other. This instability contributes to the difficulty in consistently capturing complex structures.

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4. Part 2.1: BloodMNIST (Medical Image Generation)

This section focuses on the application of Generative Adversarial Networks to medical image generation using the BloodMNIST dataset, a subset of the MedMNIST collection. The objective is to train a Deep Convolutional GAN (DCGAN) capable of generating synthetic blood cell microscope images that resemble real samples.

The approach involves initial exploration of the dataset to understand class distribution and visual characteristics, followed by the implementation and training of a convolutional GAN architecture. Model performance is evaluated using both qualitative analysis, through visual comparison of real and generated images, and quantitative indicators such as training loss behaviour. This allows for a comprehensive assessment of how effectively the model captures the underlying structure of medical image data.

4.1 Dataset and Preprocessing

The BloodMNIST dataset consists of small RGB images representing different blood cell types. Sample images from the dataset (Figure 4) demonstrate consistent structural patterns, including circular cell shapes and distinct colour variations, which provide useful visual cues for generative modelling.

Due to the original resolution (28×28), images were resized to 32×32 to ensure compatibility with convolutional architectures and to avoid shape inconsistencies during training. Pixel values were also normalised to improve training stability.

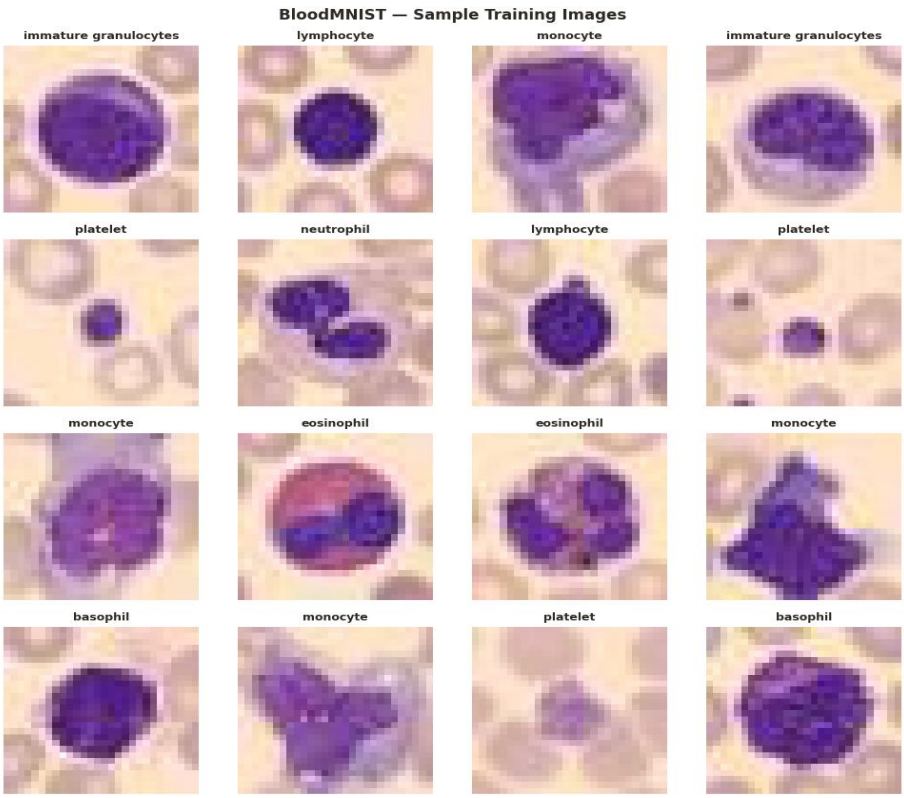

Figure 4: *Sample real BloodMNIST images illustrating variation in cell morphology and colour patterns.*

4.2 Results

The DCGAN successfully learns key visual characteristics of the dataset, including basic cellular structures, consistent colour distributions, and texture patterns resembling real images. Generated samples at later stages of training (Figure 5) show improved structural consistency and more coherent visual patterns, indicating that the model can capture spatial features through adversarial learning.

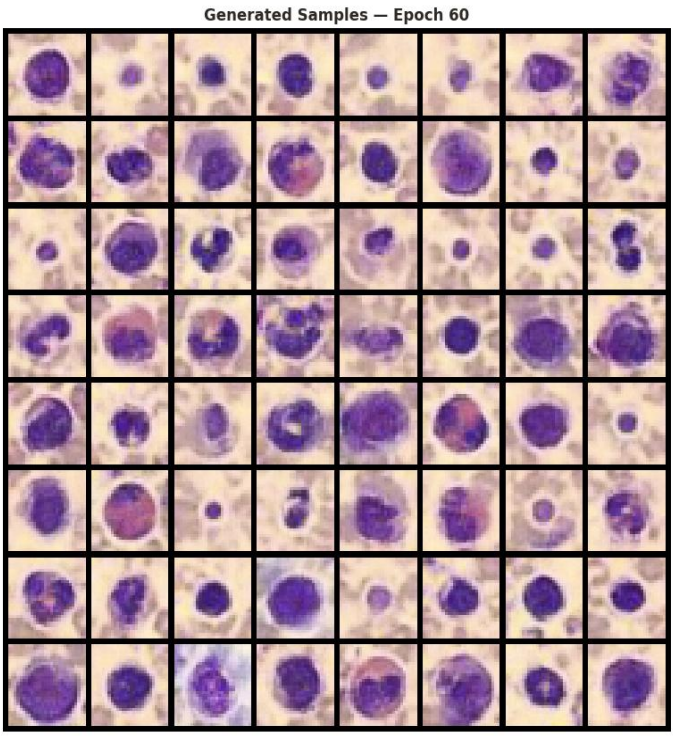

Figure 5: *Generated BloodMNIST samples at later training stages, demonstrating improved structural consistency, and more realistic visual patterns.*

4.3 Critical Analysis

Despite these strengths, several limitations are evident. Generated images lack fine-grained biological detail, and some samples exhibit blurring or repetition, indicating limited diversity. These observations are more clearly illustrated in Figure 6.

These limitations can be attributed to multiple factors. First, the low resolution of the dataset restricts the level of detail that can be learned. Second, GAN training is susceptible to **mode collapse**, where the generator produces a limited variety of outputs. Third, the instability of adversarial optimisation often leads to oscillatory learning rather than stable convergence.

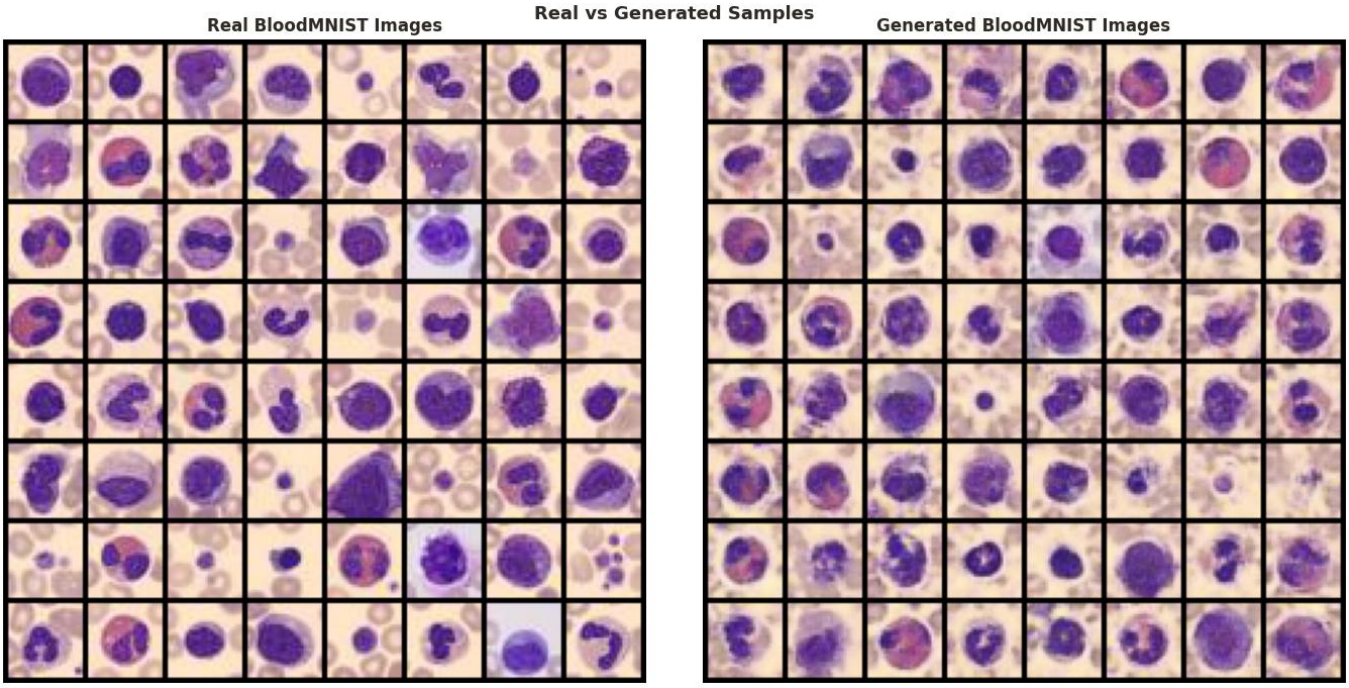

Figure 6: *Comparison of real and generated BloodMNIST images highlighting similarities in structure but reduced detail and diversity in synthetic samples.*

The observed blurring suggests that the generator prioritises capturing global structure over fine-grained detail, which is consistent with known GAN behaviour.

4.4 Training Behaviour

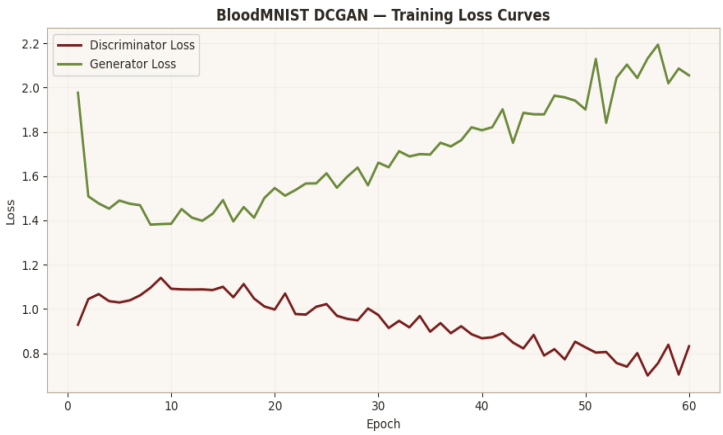

Figure 7: *Generator and discriminator loss curves illustrating oscillatory behaviour characteristic of adversarial training.*

The training dynamics provide further insight into model performance. As shown in Figure 7, both generator and discriminator losses exhibit oscillatory behaviour throughout training.

This behaviour reflects the adversarial nature of GAN optimisation, where improvements in one network can temporarily degrade the performance of the other. Unlike traditional supervised learning models, convergence is not clearly defined, and model stability must be assessed using both loss curves and visual outputs.

4.5 Insight

The results demonstrate that DCGAN is effective in capturing spatial structure and generating visually plausible images. However, this effectiveness is primarily limited to coarse structural features rather than fine-grained detail.

Importantly, in a medical context, visual realism does not imply clinical validity. Although generated images resemble real samples, there is no guarantee that they accurately represent meaningful biological patterns. This highlights a critical limitation of applying GANs in healthcare settings, where domain expertise is required to validate generated outputs.

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5. Part 2.2: Cybersecurity – Synthetic Traffic Generation (CICIDS 2017)

5.1 Dataset and Approach

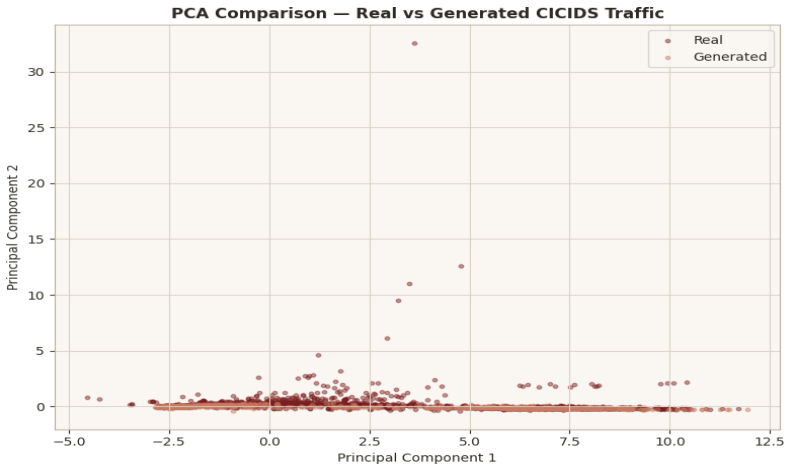
This section focuses on generating synthetic network traffic data using the CICIDS 2017 dataset. The dataset consists of high-dimensional tabular features describing network flows, including both benign and attack traffic. Unlike image data, these features do not exhibit spatial structure and instead rely on complex inter-feature relationships.

To understand the dataset, multiple files corresponding to different days were initially explored to examine feature characteristics and class balance. For modelling, the analysis was restricted to the Wednesday dataset, focusing on **BENIGN** and **DoS** traffic.

A fully connected GAN was implemented to generate synthetic feature vectors. Prior to training, the data was preprocessed through feature selection, cleaning, and scaling to ensure stable optimisation. Model performance was evaluated using both statistical comparisons and dimensionality reduction techniques, specifically Principal Component Analysis (PCA), to assess how well the generated data aligns with the real distribution.

5.2 Results

The generated samples exhibit partial alignment with the real data distribution. In PCA space, synthetic samples overlap with real data clusters, indicating that the GAN is able to capture the overall structure of the dataset.



In addition, feature-level comparisons show that the synthetic data approximates the statistical properties of key variables. However, discrepancies remain, particularly in regions where the data distribution is sparse or highly variable.

Figure 8: *PCA projection of real and generated network traffic data showing partial overlap between distributions.*

5.3 Critical Analysis

Generating tabular data using GANs presents significantly greater challenges compared to image-based tasks. This is primarily due to the absence of spatial locality, which limits the effectiveness of standard neural network architectures designed for structured data such as images.

Furthermore, tabular datasets often contain complex and nonlinear dependencies between features. These relationships are difficult to capture through implicit learning, especially when using standard GAN architectures without explicit mechanisms for modelling feature interactions.

As a result, while the model is able to capture global trends in the data distribution, it fails to reproduce precise relationships between variables. This is evident in the reduced clustering clarity observed in PCA space and the underrepresentation of certain regions of the feature space.

5.4 Insight

The results highlight a key limitation of applying standard GANs to tabular data: without incorporating mechanisms to model feature dependencies, GANs struggle to achieve high-fidelity generation in structured datasets.

This suggests that more specialised approaches, such as tabular-specific generative models or architectures designed to preserve feature relationships, may be required to improve performance in this domain.

6. Part 2.3: Creative AI – QuickDraw Sketch Generation

6.1 Dataset and Approach

This section focuses on generating synthetic sketches using the QuickDraw “pizza” dataset. The dataset consists of vector-based drawings, which were rasterised into 28×28 grayscale images to enable training with convolutional neural networks.

A Deep Convolutional GAN (DCGAN) was implemented to model the sketches. The model was trained on the rasterised images, with training progress monitored through visual outputs across epochs and loss behaviour.

6.2 Results and Evaluation

The model is able to learn basic characteristics of the sketches, including stroke density and general line structure. Generated samples at later stages of training show improved visual coherence compared to early outputs.

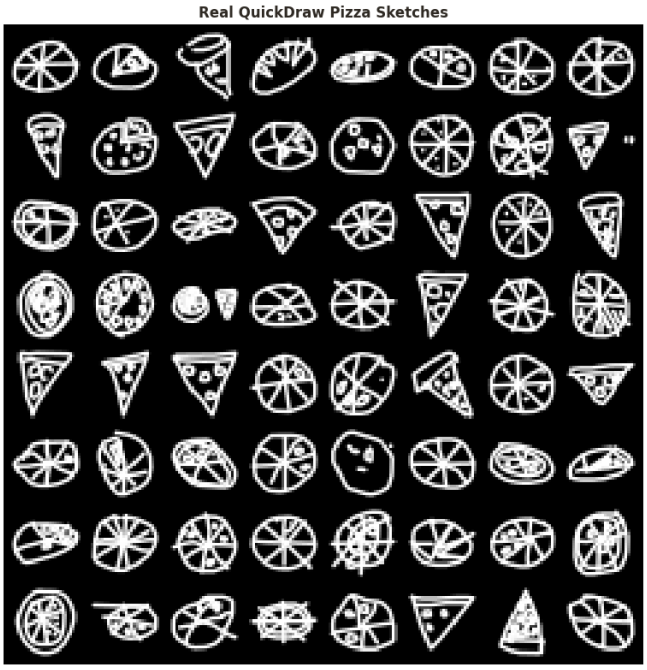


Figure 9: *Sample real QuickDraw pizza sketches illustrating high variability in drawing styles and levels of detail.*

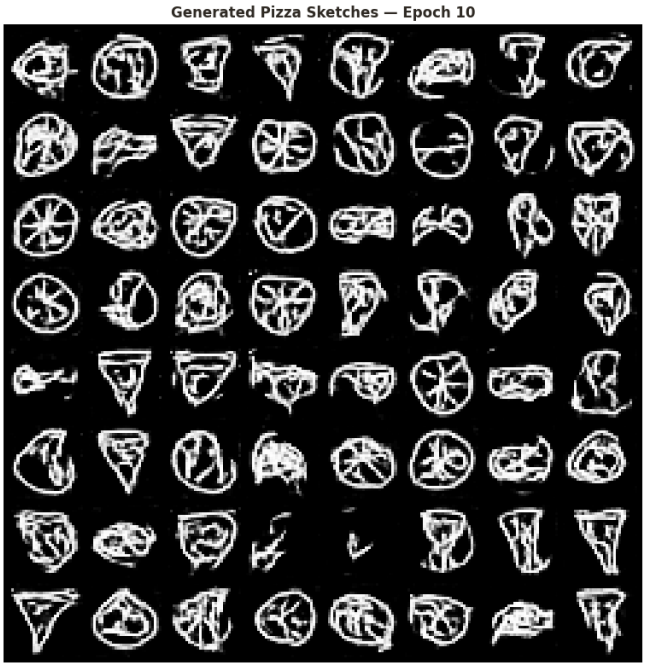


Figure 10: *Generated pizza sketches at later training stages showing improved stroke continuity and structure.*

However, many generated sketches lack clear semantic meaning, with inconsistent shapes and incomplete representations. Quantitatively, simple pixel-level comparisons indicate that the model captures general distributional properties, but these metrics do not fully reflect perceptual quality.

6.3 Critical Insight

The results highlight the difficulty of applying GANs to abstract and highly variable data. Unlike natural images, QuickDraw sketches exhibit limited structural consistency and significant stylistic variation, making it challenging for the model to learn a stable representation.

As a result, the GAN primarily captures low-level features such as strokes, but fails to reproduce higher-level semantic structure required for recognisable objects. This demonstrates that GAN performance is strongly dependent on the presence of consistent structural patterns in the data.

7. Comparative Analysis

The performance of GANs varies significantly across domains, highlighting the strong influence of data structure on generative modelling outcomes.

A key observation is that GAN performance is highly data dependent. In image-based tasks, convolutional architectures effectively exploit spatial locality, enabling the model to learn meaningful patterns. In contrast, tabular and sketch-based datasets lack consistent structural properties, making it more difficult for standard GANs to capture complex relationships.

Domain	Strength	Limitation
Synthetic	Accurate learning	Sensitive to geometry
Medical	Good structural capture	Limited fine detail
Tabular	Captures global trends	Weak feature relationships
Sketch	Learns stroke patterns	Poor semantic structure

This variation demonstrates that applying a single GAN architecture across different domains is insufficient. Instead, model design must be tailored to the characteristics of the data to achieve optimal performance.

8. Limitations

This study highlights several important limitations. First, GAN training is inherently unstable, with performance highly sensitive to architectural design and hyperparameter selection. Second, evaluation remains a challenge, as commonly used metrics do not fully capture perceptual or structural quality, particularly in non-image domains.

Additionally, the absence of domain-specific validation limits the ability to assess the real-world usefulness of generated data, especially in medical applications. Finally, dataset size and resolution constraints may have restricted the model's ability to learn more complex patterns.

9. Future Work

Future work could explore the use of conditional GANs to enable controlled data generation, as well as more advanced architectures such as Wasserstein GANs (WGANs) to improve training stability.

In addition, domain-specific generative models for tabular data may better capture feature dependencies, while higher-resolution datasets could enhance performance in image-based tasks. Incorporating domain knowledge into the modelling process may further improve the quality and reliability of generated data.

10. Conclusion

This study demonstrates that GANs are highly effective for structured data such as images, where spatial relationships can be exploited by convolutional architectures. However, their performance degrades in domains where structure is weak, abstract, or dependent on complex feature interactions.

The results highlight that generative performance is determined by the interaction between data representation, model architecture, and evaluation strategy. Consequently, GANs should not be applied as a one-size-fits-all solution but rather adapted to the specific characteristics of the problem domain.

Overall, while GANs remain a powerful generative framework, their successful application requires careful design, appropriate evaluation, and consideration of domain-specific constraints.

11. References

- Goodfellow, I. et al. (2014) *Generative Adversarial Networks*.
- Radford, A. et al. (2015) *Unsupervised Representation Learning with DCGAN*.
- MedMNIST Dataset (2023) Available at: https://medmnist.com/
- CICIDS Dataset (2017) Available at: Kaggle

Github repository link: https://github.com/imshaik7/GANs-Casestudy